

KALABE KEBEDE

Senior Software Engineer | Forward Deployed Engineer

GitHub: github.com/Kalab21

PROFESSIONAL SUMMARY

Software engineer with 6+ years of experience building and modernizing enterprise systems across banking, insurance, analytics, and AI-enabled applications. Strong in Java/Spring Boot, distributed systems, Kafka, SQL, AWS, data engineering, and production support, with recent Forward Deployed Engineering work in Python/FastAPI, RAG, AI evaluation, and brownfield modernization. Experienced moving from stakeholder requirements and system analysis through implementation, testing, observability, security, and release support.

TECHNICAL SKILLS

Software & Distributed Systems: Java 8–21, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, REST APIs, Microservices, Apache Kafka, Event-Driven Architecture, API Gateway, concurrency, idempotency

Cloud, Data & DevOps: AWS (EC2, ECS/Fargate, S3, Lambda, RDS, Glue, Athena, CloudWatch), Docker, Kubernetes, GitHub Actions, Jenkins, Maven, CI/CD

Data Engineering & Analytics: Advanced SQL, PostgreSQL, Oracle, MySQL, MongoDB, Redis, ETL/ELT, data pipelines, CDC, data quality, query optimization, indexing, data modeling, Power BI, Tableau, DAX, Power Query

AI & Forward Deployed Engineering: Python, FastAPI, RAG, LLM integration, AI evaluation, Human-in-the-Loop, brownfield analysis, requirements synthesis, spec-driven development, ADRs, client/stakeholder collaboration

Security, Reliability & Testing: OAuth2/OIDC, JWT, RBAC, OWASP concepts, secrets management, logging, metrics, tracing, reconciliation, JUnit 5, Mockito, Pytest, Playwright, integration testing, TDD

PROFESSIONAL EXPERIENCE

Revature | Remote

May 2026 – Sep 2026

Forward Deployed Engineer

- Consumer lending engagement: modernized an inherited lending platform spanning intake, identity verification, credit decisioning, manual review, disclosures, payments, servicing, and reconciliation.
- Built a grounded RAG policy assistant while keeping deterministic decisioning as the system of record; strengthened RBAC, decision finality, idempotency, auditability, and data-integrity controls.
- Translated business and regulatory requirements into specifications and ADRs; added automated verification for lending calculations and production-style logging, metrics, tracing, and CI gates.
- Skills-assessment engagement: completed candidate, trainer, reporting, and role-aware workflows across FastAPI/Next.js services; wired Nginx/API-gateway routing and PostgreSQL/MongoDB-backed features.

Lovelytics | Arlington, VA

Dec 2024 – Nov 2025

Data & Analytics Engineer — Contract

- Designed SQL-based analytics and reporting solutions using PostgreSQL/MySQL, ETL/ELT, Power BI, and dimensional modeling; translated stakeholder needs into maintainable KPI models and dashboards.
- Built AWS analytics workflows with Athena, S3, Glue, and RDS, and implemented validation and data-quality checks that improved reliability of downstream reporting.
- Optimized analytical SQL with CTEs, window functions, indexing, aggregation, and caching; automated recurring reporting and transformation workflows with Power Query and DAX.

JPMorgan Chase | Columbus, OH

May 2024 – Oct 2024

Java Software Engineer — Contract

- Modernized enterprise banking services with Java and Spring Boot, improving API/database efficiency, maintainability, and deployment reliability.
- Built secure microservices and integrations using Spring Security, JPA, Oracle, Redis, AWS, REST APIs, SFTP, and SMTP; supported issues across application, integration, and data layers.
- Improved reliability through query tuning, caching, automated tests, exception handling, Docker/Kubernetes practices, code reviews, and production support.

PROFESSIONAL EXPERIENCE (CONTINUED)

Scale AI / Outlier AI | Remote

Apr 2024 – Feb 2025

AI Data Specialist — Flexible / Task-Based Contract

- Validated and refined AI-generated datasets and responses for factual accuracy, semantic quality, and alignment with task requirements and business logic.
- Promoted to reviewer responsibilities, auditing peer work and providing targeted feedback to improve consistency and quality.

Investors Bank | Short Hills, NJ

Apr 2021 – Dec 2023

Java Software Engineer

- Modernized legacy banking workflows into Spring Boot microservices and Kafka-based event-driven services using PostgreSQL and Redis while preserving transaction and data integrity.
- Improved database and backend performance through SQL refactoring, indexing, caching, and asynchronous processing; supported production troubleshooting across distributed services.
- Automated build, test, and delivery workflows with Jenkins and GitHub Actions and collaborated with architects, QA, and business partners on incremental modernization.

Erie Insurance | Erie, PA

Mar 2019 – Feb 2021

Java Software Engineer

- Developed Java backend services and REST APIs supporting claims, policies, customers, and transaction-processing workflows.
- Implemented concurrency-safe processing, Hibernate/JPA persistence, MySQL query optimization, and automated testing for business-critical services.
- Worked with engineering and QA teams on defect resolution, production troubleshooting, reliability, maintainability, and application performance.

SELECTED ENGINEERING PROJECTS

Northbank Banking Platform — Distributed Systems & Financial Engineering | github.com/Kalab21/banking-platform

- Personal flagship platform built with Java 17, Spring Boot, Kafka, PostgreSQL, Redis, Next.js, Docker, CI/CD, and AWS-oriented infrastructure design; 11 business services plus gateway/Eureka (13 backend processes).
- Demonstrates transaction correctness, concurrency/row locking, idempotency, ownership-based authorization, loans/cards, event-driven workflows, observability, and extensive automated verification (1,482 CI tests plus live/full-stack suites).

Meridian Lending Platform — Forward Deployed Engineering & Applied AI | github.com/2463-FDE/KK-meridian-lending

- Team brownfield modernization project covering lending workflows, manual review, decision evidence, disclosures, servicing, payments, reconciliation, observability, governance, and a grounded staff policy assistant using Python/FastAPI and RAG.

Rev-Eval Evaluation Platform — AI Platform & Full-Stack FDE Engineering | github.com/RevatureFDEPEP/rev-eval

- Team multi-service assessment platform using FastAPI, Next.js, PostgreSQL, MongoDB, MinIO, Nginx, JWT/RBAC, scoring/idempotency, trainer workflows, reporting/analytics, and CI/CD.

MarketHub — Java Full-Stack Engineering | github.com/Kalab21/markethub

- Spring Boot MVC marketplace with Admin/Seller/Buyer workflows, Spring Security, JPA/MySQL, Thymeleaf, Docker, GitHub Actions, and 39 automated tests.

EDUCATION & CERTIFICATION

B.S. Computer Science — Maharishi International University — 2023

B.S. Electrical & Computer Engineering — Haramaya University — 2015

Data Analytics Certificate — Per Scholas